

Lab 4: Logistic Regression

Objective:

By the end of this lab, students should be able to implement Logistic Regression on realworld datasets using Python and scikit-learn, and practice vital machine learning skills such as data exploration and preprocessing, model evaluation, hyperparameter tuning, and cross-validation.

Software Requirements:

- (1) Python
- (2) scikit-learn pandas
- (3) numpy
- (4) matplotlib

Dataset:

You may use the ‘Titanic’ dataset provided and ‘iris’ dataset from scikit-learn. For ‘Titanic’ dataset, classifying whether passengers on the Titanic survived is a binary classification problem. For ‘iris’ dataset, you will explore multi-class classification.

Lab Instructions

1. Import necessary libraries.

2. Follow the exercises below to build and evaluate your multivariate linear regression model.

2.1 Basic exercises.

- 1) Data Loading and Exploration: (10 pts)
 - Load the dataset into a DataFrame by Pandas.
 - Use methods like ‘head()’, ‘info()’, and ‘describe()’ to explore the data.
 - Visualize distributions of key features using histograms or boxplots.
- 2) Data Preprocessing: (20 pts)
 - Split the data into training (80%) and test (20%) sets. Hint: may use ‘from sklearn.model_selection import train_test_split’
 - Handle missing values, e.g. you can use mean/median imputation, or drop rows/columns with missing values. Print the number of missing values in the processed columns **before** and **after** processing.
 - Convert categorical variables into numeric format using techniques like One-Hot Encoding, e.g. use ‘pd.get_dummies()’, or ‘OneHotEncoder’ from scikit-learn for categorical features.
 - Implement feature scaling, and print the feature you processed.

3) Model Building: (20 pts)

- Use scikit-learn's 'LogisticRegression' to create a model. (Apply L1 regularization and L2 regularization separately to build the model. And complete training and prediction on two models.)
- Train the model on the training data.
- Predict the target variable for the test data.

4) Model Evaluation (Both models need to be completed): (25 pts)

- Calculate the accuracy of the model.
- Generate a confusion matrix and visualize it.
- Calculate other metrics: precision, recall, and F1-score
- Print out the classification report. Hint: may use 'classification_report' function.
- Plot the ROC curve and calculate the AUC to evaluate the model's performance across different thresholds.

Note: You need to choose one of the models above to complete Hyperparameter Tuning. And write down which model you have chosen through markdown.

5) Hyperparameter Tuning: (15 pts)

- Use 'GridSearchCV' or 'RandomizedSearchCV' to find the best hyperparameters.
- You need to tune parameter 'C' (inverse regularization strength). (The parameter 'solver' is optional.)
- Evaluate the model with the best parameters on the test set. Compare the results of the logistic regression model before and after hyperparameter tuning through a table.

2.2 Advance exercises.

1) Multi-class Classification: (10 pts)

- Load 'iris' dataset from scikit-learn and display the first 5 rows.
- Split the data into training (80%) and test (20%) sets.
- Implement feature scaling.
- Constructing a logistic regression model using One vs Rest strategy
- Predict and evaluate the model.
- Output the accuracy, classification_report and confusion_matrix.

Submission Requirement:

- 1) Complete sections 2 in a jupyter notebook file, name your file as "lab4_student name.ipynb" and submit it on iSpace.
- 2) Please provide necessary outputs to support your assignment. (e.g. printing necessary information or visualizing through graphs and tables)

Deadline: 23:00:00, Mar.17, 2026, Plagiarism (cheating) = F